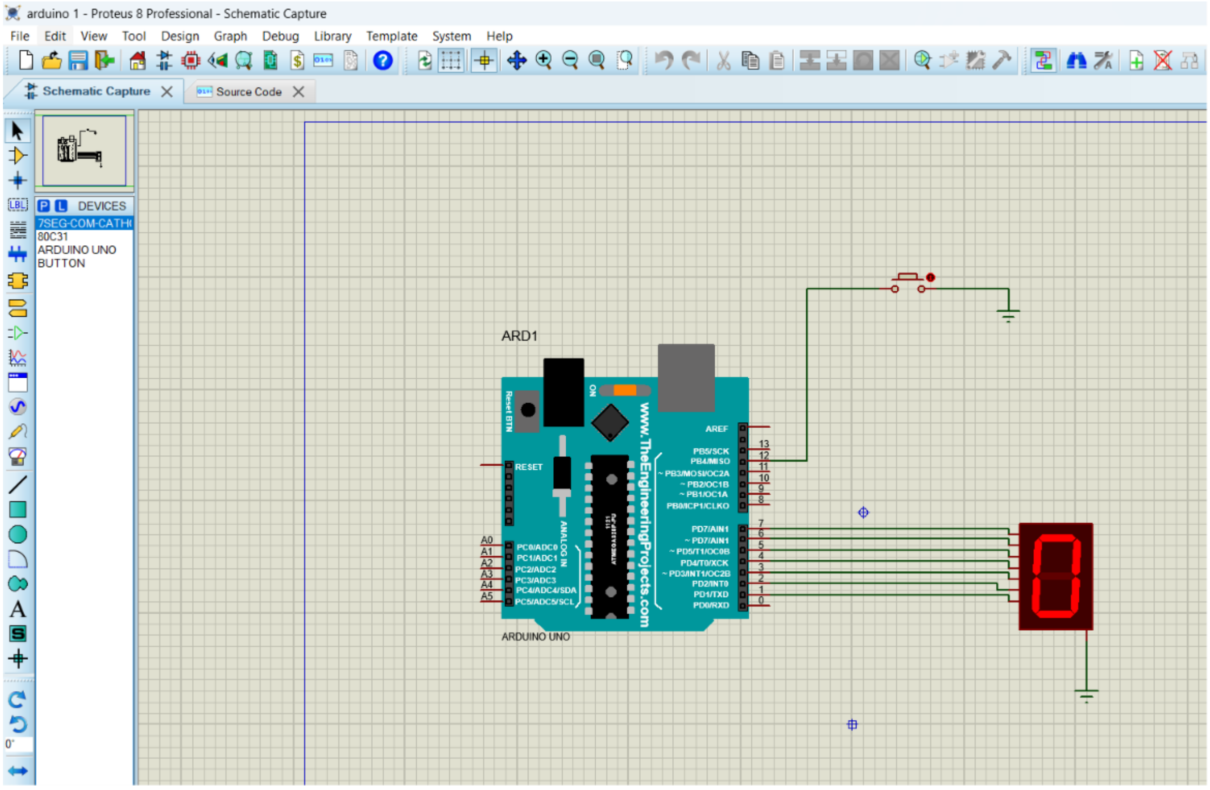
EXP -5 – ARDUINO BASED MANUAL ELECTRONIC COUNTER USING PROTEUS

CIRCUIT DIAGRAM:



CODE:

```
int x0, x1, x2, x3, x4, x5, x6, x7, x8, x9;

int delay_time = 200;


void setup() {

 // configure pin12 as an input and enable the internal pull-up resistor

 pinMode(12, INPUT_PULLUP);

 pinMode(1, OUTPUT);

 pinMode(2, OUTPUT);

 pinMode(3, OUTPUT);

 pinMode(4, OUTPUT);

 pinMode(5, OUTPUT);
```

```
 pinMode(6, OUTPUT);
 pinMode(7, OUTPUT);
}

void loop() {
 //read the pushbutton value into a variable
int sensorVal=digitalRead(12);

if(sensorVal=LOW){
 x0=true;
}
 while(x0){
 zero();
 sensorVal=digitalRead(12);
 if(sensorVal=LOW){
   x1=true;
   x0=false;
  }
   }
 while(x1){
 one();
 sensorVal=digitalRead(12);
 if(sensorVal=LOW){
   x2=true;
   x1=false;
  }
   }
 while(x2){
```

```
two();
sensorVal=digitalRead(12);
if(sensorVal=LOW){
  x3=true;
  x2=false;
 }
   }
while(x3){
three();
sensorVal=digitalRead(12);
if(sensorVal=LOW){
  x4=true;
  x3=false;
 }
   }
while(x4){
four();
sensorVal=digitalRead(12);
if(sensorVal=LOW){
  x5=true;
  x4=false;
 }
   }
while(x5){
five();
sensorVal=digitalRead(12);
if(sensorVal=LOW){
  x6=true;
```

```
  x5=false;
 }
   }
while(x6){
six();
sensorVal=digitalRead(12);
if(sensorVal=LOW){
  x7=true;
  x6=false;
 }
   }
while(x7){
seven();
sensorVal=digitalRead(12);
if(sensorVal=LOW){
  x8=true;
  x7=false;
 }
   }
while(x8){
eight();
sensorVal=digitalRead(12);
if(sensorVal=LOW){
  x9=true;
  x8=false;
 }
   }
while(x9){
```

```
 nine();
 sensorVal=digitalRead(12);
 if(sensorVal=LOW){
   x0=true;
   x9=false;
  }
   }
}

void zero()
{
 for(int i=1;i<7;i++)
 {
  digitalWrite(i,HIGH);
  digitalWrite(7,LOW);
 }
  delay(delay_time);
}

void one()
{
 digitalWrite(1,LOW);
 digitalWrite(2,HIGH);
 digitalWrite(3,HIGH);
 digitalWrite(4,LOW);
 digitalWrite(5,LOW);
 digitalWrite(6,LOW);
 digitalWrite(7,LOW);
```

```
 delay(delay_time);
}
void two()
{
 digitalWrite(1,HIGH);
 digitalWrite(2,HIGH);
 digitalWrite(3,LOW);
 digitalWrite(4,HIGH);
 digitalWrite(5,HIGH);
 digitalWrite(6,LOW);
 digitalWrite(7,HIGH);
 delay(delay_time);
}
void three()
{
 digitalWrite(1,HIGH);
 digitalWrite(2,HIGH);
 digitalWrite(3,HIGH);
 digitalWrite(4,HIGH);
 digitalWrite(5,LOW);
 digitalWrite(6,LOW);
 digitalWrite(7,HIGH);
 delay(delay_time);
}
void four()
{
 digitalWrite(1,LOW);
 digitalWrite(2,HIGH);
```

```
 digitalWrite(3,HIGH);
 digitalWrite(4,LOW);
 digitalWrite(5,LOW);
 digitalWrite(6,HIGH);
 digitalWrite(7,HIGH);
 delay(delay_time);
}
void five()
{
 digitalWrite(1,HIGH);
 digitalWrite(2,LOW);
 digitalWrite(3,HIGH);
 digitalWrite(4,HIGH);
 digitalWrite(5,LOW);
 digitalWrite(6,HIGH);
 digitalWrite(7,HIGH);
 delay(delay_time);
}
void six()
{
 digitalWrite(1,HIGH);
 digitalWrite(2,LOW);
 digitalWrite(3,HIGH);
 digitalWrite(4,HIGH);
 digitalWrite(5,HIGH);
 digitalWrite(6,HIGH);
 digitalWrite(7,HIGH);
 delay(delay_time);
```

```
}
void seven()
{
 digitalWrite(1,HIGH);
 digitalWrite(2,HIGH);
 digitalWrite(3,HIGH);
 digitalWrite(4,LOW);
 digitalWrite(5,LOW);
 digitalWrite(6,LOW);
 digitalWrite(7,LOW);
 delay(delay_time);
}
void eight()
{
 digitalWrite(1,HIGH);
 digitalWrite(2,HIGH);
 digitalWrite(3,HIGH);
 digitalWrite(4,HIGH);
 digitalWrite(5,HIGH);
 digitalWrite(6,HIGH);
 digitalWrite(7,HIGH);
 delay(delay_time);
}

void nine()
{
 digitalWrite(1,HIGH);
 digitalWrite(2,HIGH);
```

```
 digitalWrite(3,HIGH);
 digitalWrite(4,HIGH);
 digitalWrite(5,LOW);
 digitalWrite(6,HIGH);
 digitalWrite(7,HIGH);
 delay(delay_time);
}
```

OUTPUT: